



PDF Download
3511808.3557483.pdf
18 January 2026
Total Citations: 8
Total Downloads: 302

Latest updates: <https://dl.acm.org/doi/10.1145/3511808.3557483>

RESEARCH-ARTICLE

Unbiased Learning to Rank with Biased Continuous Feedback

YI REN, Tencent, Shenzhen, Guangdong, China

HONGYAN TANG, Tencent, Shenzhen, Guangdong, China

SIWEN ZHU, Tencent, Shenzhen, Guangdong, China

Open Access Support provided by:

Tencent

Published: 17 October 2022

[Citation in BibTeX format](#)

CIKM '22: The 31st ACM International
Conference on Information and
Knowledge Management
October 17 - 21, 2022
GA, Atlanta, USA

Conference Sponsors:

SIGIR
SIGWEB

Unbiased Learning to Rank with Biased Continuous Feedback

Yi Ren*
Tencent
Beijing, China
henrybjren@tencent.com

Hongyan Tang*
Tencent
Beijing, China
violatang@tencent.com

Siwen Zhu
Tencent
Beijing, China
siwenzhu@tencent.com

ABSTRACT

It is a well-known challenge to learn an unbiased ranker with biased feedback. Unbiased learning-to-rank(LTR) algorithms, which are verified to model the relative relevance accurately based on noisy feedback, are appealing candidates and have already been applied in many applications with single categorical labels, such as user click signals. Nevertheless, the existing unbiased LTR methods cannot properly handle continuous feedback, which are essential for many industrial applications, such as content recommender systems.

To provide personalized high-quality recommendation results, recommender systems need model both categorical and continuous biased feedback, such as click and dwell time. As unbiased LTR methods could not handle these continuous feedback and pair-wise learning without debiasing often performs worse than point-wise on biased feedback, which is also verified in our experiments, training multiple point-wise rankers to predict the absolute value of multiple objectives and leveraging a distinct shallow tower to estimate and alleviate the impact of position bias has been the mainstream approach in major industrial recommendation applications. However, with such a training paradigm, the optimization target differs a lot from the ranking metrics valuing the relative order of top-ranked items rather than the prediction precision of each item. Moreover, as the existing system tends to recommend more relevant items at higher positions, it is difficult for the shallow tower based methods to precisely attribute the user feedback to the impact of position or relevance. Therefore, there exists an exciting opportunity for us to get enhanced performance if we manage to solve the aforementioned issues.

Accordingly, we design a novel unbiased LTR algorithm¹ to tackle the challenges, which innovatively models position bias in the pairwise fashion and introduces the pairwise trust bias to separate the position bias, trust bias, and user relevance explicitly and can work for both continuous and categorical feedback. Experiment results on public benchmark datasets and internal live traffic of a large-scale recommender system at Tencent News show superior results for continuous labels and also competitive performance for categorical labels of the proposed method.

*Both authors contributed equally to this research.

¹The source code is released at <https://github.com/phyllist/ULTRA>.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CIKM '22, October 17–21, 2022, Atlanta, GA, USA.

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9236-5/22/10...\$15.00

<https://doi.org/10.1145/3511808.3557483>

CCS CONCEPTS

• Information systems → Learning to rank.

KEYWORDS

unbiased LTR, learning to rank, position bias, continuous feedback, recommender system, pairwise learning

ACM Reference Format:

Yi Ren, Hongyan Tang, and Siwen Zhu. 2022. Unbiased Learning to Rank with Biased Continuous Feedback. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM '22)*, October 17–21, 2022, Atlanta, GA, USA. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3511808.3557483>

1 INTRODUCTION

Currently, we see widespread adoption of unbiased learning to rank algorithms [1, 2, 10] for many scenarios with biased click signals, such as search engines. These algorithms model position bias as a counterfactual effect and estimate the unbiased relevance preference by weighting each noisy click signal with its Inverse Propensity Weight to get exciting performance gains for the corresponding applications. Nonetheless, the existing methods can only work for categorical labels instead of continuous labels, as the relationship between relevance and continuous user feedback does not adhere to their core assumptions that the binary click probability equals to the multiplication of the binary probability of relevance and examination shown in Equation 3. Moreover, as continuous feedback (e.g., dwell time and video watch ratio) is essential to accurately model user preference in many other scenarios, such as content recommender systems, it is desirable to design an unbiased LTR method for continuous labels.

Recommender systems need predict both categorical and continuous user feedback (e.g., click, purchase, share, and dwell-time) to learn user preference and recommend top-ranked items to the corresponding user. And user behaviors are often biased in recommendation scenarios. For instance, a user might click and watch a video simply because it is ranked high rather than its relevance. Thus, to estimate relative user preference accurately and improve ranking performance in recommender systems, it is essential to correct the position bias [7] from biased user behaviors. As the existing unbiased LTR methods cannot handle continuous feedback appropriately and pair-wise learning without debiasing often performs worse than point-wise on biased feedback, which is also verified by our experiments in Section 5, the mainstream industrial ranking modules usually train multiple point-wise rankers to learn the absolute probability of multiple user behaviors [25, 31] and add a shallow tower to model and eliminate position bias directly [8, 31]. However, there is a gap between the training objective of

point-wise learning and the actual ranking objective in recommendation scenarios as ranking cares about items' relative order rather than the absolute value. Moreover, bias features and relevance-related features are coupled together with the shallow tower based models, which makes it rather difficult to eliminate bias effectively [27]. Therefore, it is difficult to correct position bias and learn user preference effectively through such a learning paradigm.

To achieve effective bias correction and incorporate pair-wise learning paradigm, we propose a novel unbiased pair-wise LTR method, which models position-based examination bias in the pair-wise fashion and introduces the pairwise trust bias to better model the underlying bias and learn the relative preference more accurately. Compared with the SOTA unbiased LTR methods, our method models the correlation between the unbiased relevance pairs and the biased user feedback pairs directly, which bears no assumption mentioned above on labels and can work for both categorical and continuous labels. Moreover, pairwise debiasing is more consistent with the objective of ranking, which achieves better performance of preference learning. Furthermore, the introduction of pairwise trust bias separates the position-dependent examination bias and the position-pair-dependent relative trust bias explicitly for better correction. Offline experiment on LTR and recommendation benchmark datasets and online A/B testing in a large-scale video recommender system at Tencent News show that our method not only outperforms the SOTA ranking models for continuous labels but also performs competitively for categorical labels.

The main contributions of the paper are as follows:

- We propose a novel unbiased pairwise LTR method for better bias correction and unbiased relative preference learning. To the best of our knowledge, the proposed method is the first unbiased LTR method that can work for both categorical labels and continuous labels.
- We extend the regression-based EM algorithm [27] to the pairwise setting to estimate the parameters of the proposed method effectively.
- We conduct offline and online experiments to evaluate the performance of our method. Results on benchmark datasets show that our method achieves competitive performance for both continuous and categorical labels. Online A/B testing in a large-scale recommender system at Tencent News shows a significant 2.08% improvement of the business metric.

2 RELATED WORK

In this section, we discuss related works on unbiased learning to rank and debiasing methods in recommender systems.

2.1 Unbiased Learning to Rank Algorithms

Recently, unbiased LTR has been actively studied as a promising approach to learning from biased feedback for position bias [7] correction. Joachims [15] first present the counterfactual framework to learn the theoretically unbiased ranker via the inverse propensity weighting (IPW) estimator. To estimate the propensity, both Wang [26] and Joachims [15] propose the methods of result randomization and intervention. To reduce the negative impact on user experience, some works propose to learn the propensity from biased feedback directly, such as DLA [2], regression EM [27], and

Unbiased LambdaMART [10]. The position-dependent click noise [14] is ignored in these methods, which is addressed by [1]. However, all of the above methods can only work for categorical labels due to their core assumption on binary click probability and binary relevance probability shown in equation 3. In contrast, our proposed method can work for and achieves competitive performance for both categorical and continuous labels, according to experiment results in this paper.

2.2 Unbiased Recommender Learning from Missing-Not-At-Random Implicit Feedback

One related but different line of research [9, 18, 19, 23, 24] focuses on learning unbiased recommenders based on the Missing-Not-At-Random (MNAR) implicit feedback. These works mainly focus on eliminating exposure bias [7] and selection bias due to MNAR feedback [7] for collaborative filtering based recommendation methods [16, 17, 20]. Collaborative filtering works well for the candidate generation module in recommender systems to retrieve a set of relevant items from all candidates in low latency [11], which considers interacted items as positive and all the other items as potential negative samples but often ignores the hard negative samples. In contrast, our work targets the ranking module and adopts hard negative items that are impressed but not interacted as negative samples to rank the most desired items on the top. Moreover, our work focuses on correcting the position bias [7] rather than the selection bias and exposure bias.

2.3 Modeling Position Bias for the Ranking Module of the Recommender Systems

As mentioned above, the ranking module accepts the impressed but not interacted items as negative samples for accurate recommendation. Conventional ranking modules often adopt pointwise learning methods to estimate the interaction probability [25, 31] as unbiased LTR methods could not deal with biased continuous feedback. To correct position bias [7] for pointwise ranking models, Zhao [31] and Guo [8] both propose to employ a shallow tower to estimate and alleviate bias by combining the output of the shallow tower and the main model with summation or multiplication. However, due to the coupling of bias features and relevance-related features, shallow tower based methods cannot work effectively [27]. Wu [29] applies the idea of [27] to learn unbiased ranker with IPW framework and regression-EM for recommendation. However, it still can only work for categorical labels, which is inadequate for typical recommender systems. Our method extends the IPW framework to the pairwise setting to deal with continuous labels, which achieves superior performance of debiasing and preference learning.

3 METHODOLOGY

In this section, we provide the general framework of unbiased learning to rank for both categorical and continuous labels.

3.1 Unbiased LTR for Categorical Labels

Suppose there are N positions, for a user request u , item x_i is displayed at position $i \in [1, N]$, r_i is the unbiased relevance of user-item pair (u, x_i) . For simplicity, we only consider binary relevance,

and one can easily extend it to the multi-level case. Then we can calculate the risk function as follows:

$$R_{rel}(f) = \int L(f(u, x_i), r_i) dP(r_i = 1, u, x_i) \quad (1)$$

where $f(u, x_i)$ denotes the ranking score for user-item pair (u, x_i) , $L(f(u, x_i), r_i)$ denotes the loss function based on performance metrics and $P(r_i=1, u, x_i)$ denotes the probability distribution of positive r_i on x_i . For simplicity, the position information of items is omitted from the loss function. As most performance metrics such as Discounted Cumulative Gain (DCG) [12, 13] and Average Relevance Position (ARP) [28] only consider relevant items, the risk function here is calculated only based on items with positive relevance label.

$$\hat{f}_{rel} = \arg \min_f \sum_{u, x_i, r_i > 0} L(f(u, x_i), r_i) \quad (2)$$

If the unbiased relevance label is available, we can learn a ranker by minimizing the empirical risk function shown in Equation 2. However, it is infeasible to collect enough true relevance labels in large-scale applications as we cannot afford to find enough suitable judges for all users. Thus, abundant biased user actions (e.g., click, purchase, like) in real-world applications are valuable labels if we can fill the gap between user actions and relevance labels. As assumed in Position-Based Model (PBM) [22], the user clicks the item if and only if he examines a relevant item, and the probability of examination only depends on the position and is independent of relevance. As shown in Table 1, let e_i denotes whether item x_i is examined by the user, c_i denotes the user action label of x_i , we can model the relationship between c_i and r_i as follows:

$$\begin{aligned} P(c_i = 1|u, x_i, i) &= P(r_i = 1|u, x_i, i)P(e_i = 1|u, x_i, i) \\ &= P(r_i = 1|u, x_i)P(e_i = 1|i) \end{aligned} \quad (3)$$

Then we can derive the risk function and empirical risk function based on the user action label:

$$\begin{aligned} R_{unbiased}(f) &= \int \frac{L(f(u, x_i), c_i)}{P(e_i = 1|i)} dP(c_i = 1, u, x_i, i) \\ &= \int L(f(u, x_i), c_i) d \frac{P(c_i = 1, u, x_i, i)}{P(e_i = 1|i)} \\ &= \int L(f(u, x_i), c_i) dP(r_i = 1, u, x_i) \\ &= \int L(f(u, x_i), r_i) dP(r_i = 1, u, x_i) \\ &= R_{rel}(f) \end{aligned} \quad (4)$$

$$\hat{f}_{rel} = \hat{f}_{unbiased} = \arg \min_f \sum_{u, x_i, i, c_i > 0} \frac{L(f(u, x_i), c_i)}{P(e_i = 1|i)} \quad (5)$$

As shown in Equation 4, $R_{unbiased}$ with the inverse propensity weighting (IPW) [15] loss equals to R_{rel} in fact. In other words, if we can estimate the examination probability $P(e_i = 1|i)$ accurately, we can learn an unbiased ranker based on user action labels.

Due to the assumption between the probability distribution of relevance labels and user action labels shown in Equation 3, the aforementioned method can only be applied to tasks with categorical labels rather than continuous labels, which tends to exhibit much more complex relationships between the user action labels and the true relevance. However, continuous labels such as dwell

Notation	Description
u	User request with user profile and context.
x_i	Features of item displayed in position i .
r_i	Unbiased relevance of user-item pair (u, x_i) .
c_i	Biased feedback label of user-item pair (u, x_i) .
e_i	Examination of x_i .
θ_i	$P(e_i = 1 i)$, the probability of examination.
θ_i^-	$P(e_i = 1 i, c_i = 0)$
ϵ_{ij}^+	$P(c_i > c_j e_i = 1, e_j = 1, r_i > r_j, i, j)$
ϵ_{ij}^-	$P(c_i > c_j e_i = 1, e_j = 1, r_i \leq r_j, i, j)$
γ_{u, x_i, x_j}	$P(r_i > r_j u, x_i, x_j)$
β_{u, x_i}	$P(r_i > 0 u, x_i)$, the relevant probability of (u, x_i) .

Table 1: Notations and Descriptions

time and video watch ratio are important feedback of user satisfaction and are commonly used in industrial recommender systems. To correct position bias for continuous labels, we propose a novel unbiased LTR method which circumvents the previous assumption between unbiased relevance labels and biased user action labels. Details of the method would be provided in the following sections.

3.2 Unbiased LTR for Continuous Labels

With continuous user feedback, the aforementioned unbiased LTR method cannot work. To circumvent the limitation of the strong assumption between unbiased relevance labels and biased user action labels, we propose a novel unbiased LTR method from the perspective of pairwise learning, which can work for both categorical and continuous labels.

In the pairwise setting, the loss function is defined on item pairs. Let $\hat{y}_i = f(u, x_i)$ denotes the prediction score for user item pair (u, x_i) , $L(\hat{y}_i, r_i, \hat{y}_j, r_j)$ denotes the pairwise loss function and $P(r_i > r_j, u, x_i, x_j)$ denotes the probability distribution that user u prefers item x_i over item x_j , the risk function and the ranker learned through minimizing the empirical risk function are defined in Equation 6 and Equation 7 respectively. Similar to unbiased LTR for categorical labels, we also consider positive relevance pairs only as negative and neutral pairs should not be involved to optimize common ranking metrics such as DCG and ARP.

$$R_{rel}(f) = \int L(\hat{y}_i, r_i, \hat{y}_j, r_j) dP(r_i > r_j, u, x_i, x_j) \quad (6)$$

$$\hat{f}_{rel} = \arg \min_f \sum_{u, x_i, x_j, r_i > r_j} L(\hat{y}_i, r_i, \hat{y}_j, r_j) \quad (7)$$

3.2.1 Pairwise PBM based Unbiased LTR. To model the relation between unbiased relevance pair (r_i, r_j) and biased continuous label pair (c_i, c_j) , we extend the PBM to the pairwise setting. Specifically, we assume that the examination of item x_i and x_j are independent and if x_i and x_j are both examined, then x_i exhibits a larger continuous label than x_j if and only if x_i is more relevant than x_j , which is:

$$P(c_i > c_j, e_i = 1, e_j = 1 | u, x_i, x_j, i, j) = \theta_i \theta_j \gamma_{u, x_i, x_j} \quad (8)$$

, where $\gamma_{u,x_i,x_j} = P(r_i > r_j | u, x_i, x_j)$. Then we can derive the risk function based on the positive continuous label pairs:

$$\begin{aligned}
 R_{unbiased}(f) &= \int \frac{L(\hat{y}_i, c_i, \hat{y}_j, c_j)}{\theta_i \theta_j} P(e_j = 1 | u, i, j, x_i, x_j, c_i > c_j) dP(c_i > c_j, u, x_i, x_j, i, j) \\
 &= \int \frac{L(\hat{y}_i, c_i, \hat{y}_j, c_j)}{\theta_i \theta_j} dP(c_i > c_j, u, x_i, x_j, i, j, e_i = 1, e_j = 1) \\
 &= \int \frac{L(\hat{y}_i, c_i, \hat{y}_j, c_j)}{\theta_i \theta_j} dP(r_i > r_j, u, x_i, x_j, i, j, e_i = 1, e_j = 1) \\
 &= \int \frac{L(\hat{y}_i, r_i, \hat{y}_j, r_j)}{\theta_i \theta_j} dP(r_i > r_j, u, x_i, x_j, i, j) P(e_i = 1 | i) P(e_j = 1 | j) \\
 &= \int L(\hat{y}_i, r_i, \hat{y}_j, r_j) dP(r_i > r_j, u, x_i, x_j) = R_{rel}(f) \quad (9)
 \end{aligned}$$

Based on the justification that $R_{unbiased}$ is an unbiased estimator of R_{rel} , we can learn an unbiased ranker through minimizing the empirical risk function $loss_{IPW}$ shown in Equation 10, where $L_{ij} = L(\hat{y}_i, c_i, \hat{y}_j, c_j)$. It is worth noting that samples with positive labels and zero labels are separately handled explicitly for better precision in the empirical risk function, as the posterior probability of examination is different for samples with different labels. In detail, the posterior examination probability is definitely 1 for items with a positive label but is uncertain for items with zero labels.

$$loss_{IPW} = \sum_{u, x_i, x_j, i, j, c_i > c_j, c_j > 0} \frac{L_{ij}}{\theta_i \theta_j} + \sum_{u, x_i, x_j, i, j, c_i > c_j, c_j = 0} \frac{L_{ij} h_{ij}}{\theta_i \theta_j}, \quad (10)$$

where $h_{ij} = P(e_j = 1 | u, x_i, x_j, i, j, c_i > c_j, c_j = 0)$.

3.2.2 Pairwise Trust Bias based Unbiased LTR. The pairwise PBM based unbiased LTR method assumes that if x_i is more relevant than x_j and they are both examined, x_i will exhibit a larger continuous label than x_j . However, according to the user study in [14], users tend to trust higher-ranked items more, so higher-ranked items exhibit larger click ratios even if items in different positions are all examined and equally relevant, which is called the trust bias. Moreover, the continuous feedback value usually follows a distribution rather than a specific mapping from the relevance label, which engenders even more complex relations between biased label pairs and unbiased relevance pairs. To estimate the relation between unbiased relevance pair (r_i, r_j) and biased continuous label pair (c_i, c_j) more accurately, we incorporate the trust bias explained above and extend it to the pairwise setting in this paper. Since the trust bias only depends on the exposed position, we formulate the pairwise trust bias in Equation 11 and Equation 12 and assume that $P(c_i > c_j | e_i = 1, e_j = 1, r_i > r_j, u, x_i, x_j, i, j)$ and $P(c_i > c_j | e_i = 1, e_j = 1, r_i \leq r_j, u, x_i, x_j, i, j)$ are independent of (u, x_i, x_j) . Specifically, for a positive relevance pair $r_i > r_j$, the probability of exposing a positive continuous label pair $c_i > c_j$ is not 1 but ϵ_{ij}^+ , while a non-positive relevance pair also has the probability ϵ_{ij}^- to exhibit a positive continuous label pair. ϵ_{ij}^+ and ϵ_{ij}^- are only position-dependent and follows the constraint $0 < \epsilon_{ij}^- < \epsilon_{ij}^+ < 1$.

$$P(c_i > c_j | e_i = 1, e_j = 1, r_i > r_j, i, j) = \epsilon_{ij}^+ \quad (11)$$

$$P(c_i > c_j | e_i = 1, e_j = 1, r_i \leq r_j, i, j) = \epsilon_{ij}^- \quad (12)$$

Then we can calculate the probability of a positive continuous label pair. As shown in Equation 13, there are two cases to expose a positive continuous label pair $c_i > c_j$. First, if x_j is examined, the relation between c_i and c_j depends on the relation between r_i and r_j and the trust bias. Second, if x_j is not examined, the positive pair $c_i > c_j$ reduces to the positive data point $c_i > 0$, which only depends on the examination and relevance of x_i according to PBM.

$$\begin{aligned}
 P(c_i > c_j | u, x_i, x_j, i, j) &= P(c_i > c_j, e_i = 1, e_j = 1 | u, x_i, x_j, i, j) \\
 &\quad + P(c_i > 0, e_i = 1, e_j = 0 | u, x_i, x_j, i, j) \\
 &= \theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i} \quad (13)
 \end{aligned}$$

, where $\beta_{u, x_i} = P(r_i > 0 | u, x_i)$ is the relevant probability of (u, x_i) .

As we can no longer deduce the state of $r_i > r_j$ from the condition of $c_i > c_j, e_i = 1, e_j = 1$, R_{rel} could not be derived from $R_{unbiased}$ with the introduction of pairwise trust bias. To tackle this issue, we calculate the probability of positive relevance pair given the positive continuous label pair and true examination, which is:

$$\begin{aligned}
 P(r_i > r_j | u, x_i, x_j, i, j, c_i > c_j, e_i = 1, e_j = 1) &= \frac{P(c_i > c_j | e_i = 1, e_j = 1, r_i > r_j, u, x_i, x_j, i, j) P(r_i > r_j | u, x_i, x_j)}{P(c_i > c_j, e_i = 1, e_j = 1 | u, x_i, x_j, i, j)} \\
 &= \frac{\epsilon_{ij}^+ \gamma_{u, x_i, x_j}}{\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})} = m_{ij} \quad (14)
 \end{aligned}$$

Finally, we define a Bayes-IPW loss based on $loss_{IPW}$ and m_{ij} , which compensates and corrects for both the position-based examination bias and trust bias. One can see from Equation 15 and Equation 14 that the Bayes-IPW loss reduces to the $loss_{IPW}$ in the noisy free case of pairwise PBM, i.e., when $\epsilon_{ij}^+ = 1$ and $\epsilon_{ij}^- = 0$.

$$loss_{Bayes-IPW} = \arg \min \sum_{\substack{f_{u, x_i, x_j, i, j} \\ c_i > c_j, c_j > 0}} \frac{L_{ij} m_{ij}}{\theta_i \theta_j} + \sum_{\substack{f_{u, x_i, x_j, i, j} \\ c_i > c_j, c_j = 0}} \frac{L_{ij} m_{ij} h_{ij}}{\theta_i \theta_j}, \quad (15)$$

where h_{ij} is calculated as follows:

$$\begin{aligned}
 h_{ij} &= P(e_j = 1 | u, x_i, x_j, i, j, c_i > c_j, c_j = 0) \\
 &= \frac{P(e_j = 1 | u, x_j, j, c_j = 0) P(c_i > c_j | u, x_i, x_j, i, j, e_j = 1, c_j = 0)}{P(c_i > c_j | u, x_i, x_j, i, j, c_j = 0)} \\
 &= \frac{\theta_j \theta_j^- (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j}))}{\theta_j \theta_j^- (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_j (1 - \theta_j^-) \beta_{u, x_i}}, \quad (16)
 \end{aligned}$$

where θ_j^- denotes the posterior examination probability when the continuous label of item x_j is known to be 0.

3.2.3 Direct Optimization for Ranking Metrics. As pairwise loss [4] focuses on minimizing the number of pairwise errors and neglecting the relative importance of item pairs with different relevance gaps, which does not match well with common performance metrics such as DCG and ARP. To directly optimize the ranking metrics in the model, we refine the loss function following the practice of LambdaRank [3]. The main idea of LambdaRank is to incorporate a delta NDCG to directly optimize the evaluation metric of NDCG, where delta NDCG denotes the difference between NDCG scores if item x_i and x_j are swapped in the ranking list [3, 5]. As the unbiased true relevance is not available, we assume that the biased feedback

$$\begin{aligned}
P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i > c_j) &= \frac{\theta_i \theta_j \epsilon_{ij}^+ \gamma_{u, x_i, x_j}}{\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i}} \\
P(e_i = 1, e_j = 1, r_i \leq r_j | u, x_i, x_j, i, j, c_i > c_j) &= \frac{\theta_i \theta_j \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})}{\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i}} \\
P(e_i = 1, e_j = 0, r_i > 0 | u, x_i, x_j, i, j, c_i > c_j) &= \frac{\theta_i (1 - \theta_j) \beta_{u, x_i}}{\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i}} \\
P(e_i = 1, e_j = 1, r_i \leq r_j | u, x_i, x_j, i, j, c_i \leq c_j) &= \frac{\theta_i \theta_j (1 - \epsilon_{ij}^-) (1 - \gamma_{u, x_i, x_j})}{1 - (\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i})} \\
P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i \leq c_j) &= \frac{\theta_i \theta_j (1 - \epsilon_{ij}^+) \gamma_{u, x_i, x_j}}{1 - (\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i})} \\
P(e_i = 1, e_j = 0, r_i \leq 0 | u, x_i, x_j, i, j, c_i > c_j) &= 0 \quad P(e_i = 0 | u, x_i, x_j, i, j, c_i > c_j) = 0 \\
P(e_i = 0, r_i > 0 | u, x_i, i, c_i = 0) &= \frac{(1 - \theta_i) \beta_{u, x_i}}{1 - \theta_i \beta_{u, x_i}} \quad P(e_i = 1, r_i > 0 | u, x_i, i, c_i = 0) = 0 \\
P(e_i = 1, r_i \leq 0 | u, x_i, i, c_i = 0) &= \frac{\theta_i (1 - \beta_{u, x_i})}{1 - \theta_i \beta_{u, x_i}} \quad P(e_i = 1, r_i > 0 | u, x_i, i, c_i > 0) = 1
\end{aligned}$$

Figure 1: Formulas in Expectation Step

is positively correlated to the unbiased relevance, and refine the pairwise loss $loss_{Bayes-IPW}$ to $loss_{opt}$ as follows:

$$loss_{opt} = \sum_{\substack{u, x_i, x_j, i, j, \\ c_i > c_j, c_j > 0}} \frac{L_{ij} m_{ij} |\Delta Z_{ij}|}{\theta_i \theta_j} + \sum_{\substack{u, x_i, x_j, i, j, \\ c_i > c_j, c_j = 0}} \frac{L_{ij} m_{ij} h_{ij} |\Delta Z_{ij}|}{\theta_i \theta_j}, \quad (17)$$

where ΔZ_{ij} is the difference between evaluation metrics based on the biased label if item x_i and x_j are swapped in the ranking list.

Thus, if we can estimate the parameters of θ_i , θ_j^- , ϵ_{ij}^+ and ϵ_{ij}^- , we can learn an unbiased ranker $\hat{f}_{unbiased}$ through minimizing the loss function. In this paper, we employ a regression-based Expectation-Maximization (EM) method [27] to estimate the parameters. The estimation process would be described in the next section.

3.3 Supervision Label Synthesis for Multi-Objective Applications

Some applications, such as content recommender systems, often consider multiple live metrics simultaneously [25]. For example, we would like to maximize the user's video view count, dwell time, and other satisfaction-related metrics (e.g., liking and sharing) in the video recommendation scenario. To model multiple actions and maximize multiple live metrics simultaneously, conventional recommender systems often train multiple rankers to predict the absolute value of multiple objectives and combine these predictions for final ranking. However, the optimization target in such a training paradigm differs a lot from the ranking metrics valuing at the relative order of top-ranked items rather the absolute scores. Moreover, if we train multiple rankers through different unbiased LTR models, we could not combine predictions in a theoretically sound way as predictions of different unbiased LTR models might not be in the same scale.

To address these issues and achieve better multi-objective optimization performance, we propose to define a synthesized label based on diverse user actions so as to directly model multiple live metrics in one single model. As shown in Equation 18, the synthesized label merges multiple user actions (e.g., click, dwell time, and

satisfaction) through a combination function Φ , where M is the number of user actions considered in live metrics. The combination function is usually set based on business goals, and its form can be fairly flexible to capture complex relations between live metrics. For instance, weighted summation and weighted multiplication are commonly used in industrial recommender systems. With the synthesized continuous label, we can naturally apply the method described in section 3.2.

$$c = \Phi(action_1, action_2, ..., action_M) \quad (18)$$

There are some advantages to combining multiple user behaviors to the synthesized label. First, the gap between the training objective and the live metrics can be bridged through learning from the synthesized label, as the synthesized label is closely related to the ultimate business goals. Second, we can obtain the unbiased prediction of the live metric from the single model and employ it for final ranking directly without combination, which is more convenient and efficient.

4 ESTIMATION VIA EXPECTATION-MAXIMIZATION

To estimate parameters of θ_i , θ_j^- , ϵ_{ij}^+ and ϵ_{ij}^- introduced in the last section and learn the unbiased ranker, we extend the regression-based EM method [27] to the pairwise setting in this paper. In the procedure of regression-based EM algorithm, parameters are estimated by iterating over the Expectation steps and Maximization steps until convergence. In this section, we illustrate the process of the Expectation step and the Maximization step respectively.

4.1 Expectation Step

In the Expectation step, we need to estimate the distribution of hidden variable e_i , e_j and relevance pair (r_i, r_j) given parameters θ_i , ϵ_{ij}^+ and ϵ_{ij}^- and γ_{u, x_i, x_j} . To achieve the goal, we first calculate the joint probability of all hidden variables shown in Figure 1, where all formulas are derived directly from Bayes rules. For instance, in

$$\begin{aligned}
\epsilon_{ij}^+ &= \frac{\sum_{u, x_i, x_j, c_i > c_j} P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i > c_j)}{\sum_{u, x_i, x_j, i, j, c_i > c_j} P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i > c_j) + \sum_{u, x_i, x_j, i, j, c_i \leq c_j} P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i \leq c_j)} \\
\epsilon_{ij}^- &= \frac{\sum_{u, x_i, x_j, c_i > c_j} P(e_i = 1, e_j = 1, r_i \leq r_j | u, x_i, x_j, i, j, c_i > c_j)}{\sum_{u, x_i, x_j, i, j, c_i > c_j} P(e_i = 1, e_j = 1, r_i \leq r_j | u, x_i, x_j, i, j, c_i > c_j) + \sum_{u, x_i, x_j, i, j, c_i \leq c_j} P(e_i = 1, e_j = 1, r_i \leq r_j | u, x_i, x_j, i, j, c_i \leq c_j)} \\
\gamma_{u, x_i, x_j} &= \frac{\sum_{u' = u, x'_i = x_i, x'_j = x_j, c_i > c_j} P(r_i > r_j | u, x_i, x_j, i, j, c_i > c_j) + \sum_{u' = u, x'_i = x_i, x'_j = x_j, c_i \leq c_j} P(r_i > r_j | u, x_i, x_j, i, j, c_i \leq c_j)}{\sum_{u' = u, x'_i = x_i, x'_j = x_j} 1} \\
\beta_{u, x_i} &= \frac{\sum_{u' = u, x'_i = x_i, c_i > 0} P(r_i > 0 | u, x_i, i, c_i > 0) + \sum_{u' = u, x'_i = x_i, c_i \leq 0} P(r_i > 0 | u, x_i, i, c_i \leq 0)}{\sum_{u' = u, x'_i = x_i} 1} \\
\theta_k^- &= \frac{\sum_{u, x_i, i, c_i = 0, i=k} P(e_i = 1 | u, x_i, i, c_i = 0)}{\sum_{u, x_i, i, c_i = 0} 1\{i = k\}} \quad \theta_k = \frac{\sum_{u, x_i, i, c_i > 0, i=k} P(e_i = 1 | u, x_i, i, c_i > 0) + \sum_{u, x_i, i, c_i = 0, i=k} \theta_k^-}{\sum_{u, x_i, i} 1\{i = k\}}
\end{aligned}$$

Figure 2: Formulas in Maximization Step

the first equation, we have:

$$\begin{aligned}
&P(e_i = 1, e_j = 1, r_i > r_j | u, x_i, x_j, i, j, c_i > c_j) \\
&= \frac{P(c_i > c_j | e_i = 1, e_j = 1, r_i > r_j) P(e_i = 1, e_j = 1, r_i > r_j)}{P(c_i > c_j | u, x_i, x_j, i, j)} \\
&= \frac{\theta_i \theta_j \epsilon_{ij}^+ \gamma_{u, x_i, x_j}}{\theta_i \theta_j (\epsilon_{ij}^+ \gamma_{u, x_i, x_j} + \epsilon_{ij}^- (1 - \gamma_{u, x_i, x_j})) + \theta_i (1 - \theta_j) \beta_{u, x_i}} \quad (19)
\end{aligned}$$

It is worth noting that we estimate hidden variables e_i , e_j and true relevance $r_i > 0$ in the pointwise fashion, which is shown in the last 4 equations in Figure 1, as they only depend on a single item. After calculating the joint probability, we can then calculate the marginal probability of hidden variables, which will be used in the Maximization step. For instance, the marginal probability $P(e_i = 1 | c_i = 0)$ equals to the sum of $P(e_i = 1, r_i > 0 | c_i = 0)$ and $P(e_i = 1, r_i \leq 0 | c_i = 0)$.

4.2 Maximization Step

In the maximization step, all parameters are updated to minimize the loss function, given the training samples and the posterior probabilities from the Expectation step. All formulas for parameter updating are presented in Figure 2. In the procedure of standard EM algorithm, γ_{u, x_i, x_j} and β_{u, x_i} are updated following the 3rd equation and 4th equation respectively, which requires that (u, x_i, x_j) should repeat and be shown in different position pairs. As samples of user-item-pair (u, x_i, x_j) are highly sparse and synthesized label pairs (c_i, c_j) are noisy in large-scale real-world systems, it is extremely difficult to minimize loss through free parameters γ_{u, x_i, x_j} and β_{u, x_i} . To address the problem, we apply the regression-based EM [27] algorithm to estimate parameters of γ_{u, x_i, x_j} and β_{u, x_i} via learning a regression function. Specifically, we assume that there are feature vector X_{u, x_i, x_j} and X_{u, x_i} representing the sample of user-item-pair (u, x_i, x_j) and user-item (u, x_i) respectively, we use function g to compute the relevance preference $\gamma_{u, x_i, x_j} = g(X_{u, x_i, x_j})$ and function h to compute absolute relevance $\beta_{u, x_i} = h(X_{u, x_i})$. Thus, we aim to find regression functions $g(X)$ and $h(X)$ in the maximization step to minimize the loss function, given training samples and the distributions of hidden variables calculated in the Expectation step. For instance, we can regress the feature vector X_{u, x_i, x_j} to the probability $P(r_i > r_j | u, x_i, x_j, i, j, c_i, c_j)$. Similar to [27], we convert such a regression problem to a classification problem based on

Bernoulli sampling. Specifically, we sample a binary relevance pair label $\gamma \in \{0, 1\}$ and a binary relevance label $\beta \in \{0, 1\}$ according to $P(r_i > r_j | u, x_i, x_j, i, j, c_i, c_j)$ and $P(r_i > 0 | u, x_i, i, c_i)$ respectively. Then we can adopt classification models to learn $g(X)$ and $h(X)$ based on training set $\{X, \gamma\}$ and $\{X, \beta\}$. It is flexible to use any classification model for $g(X)$ and $h(X)$ and we choose the commonly used DNN model in this paper.

Note that the original EM algorithm updates parameters in the Maximization step through calculation on the whole data set, which imposes challenges for industrial systems with large-scale training data. To address the problem, we employ the mini-batch EM following the idea of online EM [6]. Accordingly, we calculate the formulas in Figure 2 based on data in one single mini-batch, and update parameters incrementally in each batch as follows:

$$\epsilon_{ij}^+ = \epsilon_{ij}^+ \times (1 - \alpha) + \widehat{\epsilon}_{ij}^+ \times \alpha \quad (20)$$

where α is the scheduled learning rate of ϵ_{ij}^+ , $\widehat{\epsilon}_{ij}^+$ is the estimation of ϵ_{ij}^+ in the current batch.

5 EXPERIMENTS

In this section, we first describe the experiment data and experiment setup, then conduct extensive offline and online experiments to answer the following research questions:

- RQ1: How does the proposed method perform compared with the SOTA debiasing methods on continuous labels?
- RQ2: How does the proposed method perform compared with the SOTA unbiased LTR methods on categorical labels?
- RQ3: Whether the proposed method is robust in different severity of position bias and different relative weight of continuous labels over categorical labels?
- RQ4: How does the proposed method perform in real-world recommender systems?

5.1 Experiment Data

To answer the aforementioned research questions and evaluate the effectiveness of the proposed method in both recommendation and searching scenarios, we conduct experiments on both LTR and recommendation public benchmark datasets, which are:

	Yahoo Dataset			MovieLens		
	Reward@3	Reward@5	Reward@10	Reward@3	Reward@5	Reward@10
Our Method ($loss_{opt}$)	2.3319 $\pm$ 6.9e-3	2.7185 $\pm$ 7.5e-3	2.9735 $\pm$ 5.9e-3	5.8553 $\pm$ 1.1e-2	7.6690 $\pm$ 1.5e-2	9.1400 $\pm$ 1.6e-2
Our Method ($loss_{Bayes-IPW}$)	2.3046 $\pm$ 8.5e-3	2.6864 $\pm$ 7.5e-3	2.9423 $\pm$ 8.0e-3	5.8415 $\pm$ 1.4e-2	7.6486 $\pm$ 7.8e-3	9.1217 $\pm$ 9.7e-3
Our Method ($loss_{IPW}$)	2.2447 $\pm$ 9.9e-3	2.6239 $\pm$ 1.0e-2	2.8828 $\pm$ 8.9e-3	5.8584 $\pm$ 2.0e-2	7.6635 $\pm$ 2.1e-2	9.1285 $\pm$ 2.5e-2
Sum-Synthesize	2.2599 $\pm$ 5.0e-3	2.6380 $\pm$ 8.1e-3	2.8913 $\pm$ 8.7e-3	5.7863 $\pm$ 2.5e-2	7.5704 $\pm$ 3.1e-2	9.0289 $\pm$ 3.5e-2
PAL-Synthesize	2.2612 $\pm$ 8.7e-3	2.6388 $\pm$ 6.0e-3	2.8914 $\pm$ 5.6e-3	5.7484 $\pm$ 3.1e-2	7.5271 $\pm$ 3.7e-2	8.9682 $\pm$ 4.0e-2
Sum-Click	2.2776 $\pm$ 1.2e-2	2.6590 $\pm$ 9.7e-3	2.9146 $\pm$ 8.9e-3	5.7283 $\pm$ 2.1e-2	7.5570 $\pm$ 2.3e-2	9.0264 $\pm$ 2.8e-2
PAL-Click	2.2907 $\pm$ 9.9e-3	2.6719 $\pm$ 6.6e-3	2.9274 $\pm$ 6.7e-3	5.7385 $\pm$ 1.9e-2	7.5838 $\pm$ 1.8e-2	9.0586 $\pm$ 1.9e-2
PAL-Click + PAL-DwellTime	2.2707 $\pm$ 7.7e-3	2.6504 $\pm$ 7.3e-3	2.9058 $\pm$ 6.9e-3	5.7569 $\pm$ 2.7e-2	7.5491 $\pm$ 3.5e-2	9.0031 $\pm$ 3.7e-2
Lower Bound (pointwise DNN)	2.2548 $\pm$ 4.4e-3	2.6353 $\pm$ 6.5e-3	2.8892 $\pm$ 5.3e-3	5.7106 $\pm$ 1.3e-2	7.5392 $\pm$ 1.6e-2	9.0113 $\pm$ 1.9e-2
Lower Bound (LambdaRank)	2.1619 $\pm$ 1.2e-2	2.5370 $\pm$ 8.5e-3	2.7957 $\pm$ 8.4e-3	5.6800 $\pm$ 2.1e-2	7.4805 $\pm$ 2.9e-2	8.9387 $\pm$ 4.7e-2

Table 2: Experiment Results on Continuous Labels

- LTR dataset: Yahoo! LETOR set ¹² and MSLR-Web30K ³ are two of the largest public LTR benchmark datasets, which contain multiple query document pairs represented by feature vectors and 5-level relevance labels. Specifically, there are 29.9k queries with 710k documents in Yahoo Dataset and 31.2k queries in MSLR-Web30K. We adopt the training, validation, testing splits in these corpora.
- Recommendation dataset: MovieLens ⁴ is a recommendation benchmark dataset containing 1M unbiased ratings of 6k users on 3.9k movies. As each user and movie is only represented by an id without other information in MovieLens, we generate feature vectors for all user movie pairs in the following steps. First, we split samples of each user into training(50%), validation(25%), and testing(25%) set based on the time order. Then we train a BPR [20] model based on the training set to obtain a 60-dimensional feature vector for each user and each item. Finally, we concatenate the feature vector of user, item, and dot of user and item to generate a 180-dimensional feature vector for each user-item pair.

As there is no user action in these datasets, we consider each query or user as a user request and generate the click and dwell-time labels to simulate biased feedback.

5.1.1 Click Simulation. We follow the procedures in [2] to sample clicks. First, we train a Ranking-SVM ⁵ model using 1% of the training data with relevance labels to generate the initial ranking list for each user. Then we sample click label $click_{u,x_i,i}$ uniformly by simulating the browsing process of PBM as follows:

$$click_{u,x_i,i} \sim U(P(c_i = 1|u, x_i, i)),$$

$$P(c_i = 1|u, x_i, i) = P(e_i = 1|i)P(r_i = 1|u, x_i), \quad (21)$$

where $P(e_i = 1|i) = \theta_i^\eta$ denotes the probability of examination, θ_i is the position bias of click at position i determined by the initial Ranking-SVM ranker and η controls the severity of position bias. θ_i is obtained from an eye-tracking experiment in [14] and η is set as 1 in default. The probability of an item x_i to be perceived as relevant is calculated as:

²<https://webscope.sandbox.yahoo.com/catalog.php?datatype=c>

³<https://www.microsoft.com/en-us/research/project/mslr/>

⁴<https://files.grouplens.org/datasets/movielens/ml-1m-README.txt>

⁵http://www.cs.cornell.edu/people/tj/svm_light/svm_rank.html

$$P(r_i = 1|u, x_i) = \epsilon + (1 - \epsilon) \frac{2^{y_{u,x_i}} - 1}{2^{y_{max}} - 1},$$

where $y_{u,x_i} \in [0, 4]$ is the 5-level relevance label and y_{max} is the highest level of 4. We map ratings in MovieLens from [1-5] to [0-4] as relevance labels. ϵ models click noise so that irrelevant items ($y_{u,x_i}=0$) have non-zero click probability. ϵ is set to 0.1 by default.

5.1.2 Dwell-Time Simulation. Then we sample continuous dwell-time labels based on sampled click data as only clicked samples have non-zero dwell-time in practice. As shown in [30], the dwell-time is correlated with the relevance and item length, and dwell-time of items with the same length satisfies a log-Gaussian distribution. So we analyze dwell-time of videos around 100s in a real-world video recommender system and also find a log-Gaussian distribution. Furthermore, we assume that dwell-time in different relevance levels follows different log-Gaussian distribution and the overall log-Gaussian of dwell-time is composed of these distributions. Then we train a GMM(Gaussian Mixture Model) [21] on log scaled dwell-time to obtain 5 Gaussian distributions for different relevance level and sample dwell-time as follows:

$$dt_{u,x_i,i} = click_{u,x_i,i} \cdot e^{\omega_{u,x_i}}, \quad \omega_{u,x_i} \sim N(\mu_{y_{u,x_i}}, \sigma_{y_{u,x_i}}), \quad (22)$$

where $click_{u,x_i,i}$ is the sampled click label, ω_{u,x_i} denotes the sampled log-scaled dwell-time after click, $\mu_{y_{u,x_i}}$ and $\sigma_{y_{u,x_i}}$ are mean and standard deviation in different relevance level obtained by GMM.

5.2 Experiment Setup

In this paper, we perform experiments on the synthesized label in Subsection 5.3 and the click label in Subsection 5.4 to evaluate the performance of our method on continuous and categorical labels respectively. Based on Subsection 3.3, we adopt the commonly used weighted summation function in typical recommendation scenarios and define the synthesized label $c_{u,x_i,i}$ as follows:

$$c_{u,x_i,i} = click_{u,x_i,i} + \frac{dt_{u,x_i,i}}{e^\delta},$$

where δ controls the weight of continuous label in the synthesized label. δ is set to 3 by default to keep the similar weight of dwell-time as our real-world recommender system.

In the experiment, we adopt a DNN model with [512, 256, 128] hidden units, ELU activation, Adagrad optimizer, cross-entropy loss

	Yahoo Dataset			MSLR-Web30K		
	NDCG@3	NDCG@5	NDCG@10	NDCG@3	NDCG@5	NDCG@10
Upper Bound	0.6960 $\pm$ 1.4e-3	0.7175 $\pm$ 1.3e-3	0.7644 $\pm$ 1.0e-3	0.3960 $\pm$ 2.3e-3	0.4047 $\pm$ 1.8e-3	0.4286 $\pm$ 1.8e-3
Our Method ($loss_{opt}$)	0.6895 $\pm$ 2.5e-3	0.7115 $\pm$ 2.1e-3	0.7589 $\pm$ 1.4e-3	0.3811 $\pm$ 2.6e-3	0.3899 $\pm$ 1.7e-3	0.4132 $\pm$ 1.0e-3
Our Method ($loss_{Bayes-IPW}$)	0.6900 $\pm$ 2.4e-3	0.7116 $\pm$ 1.7e-3	0.7589 $\pm$ 1.5e-3	0.3800 $\pm$ 2.5e-3	0.3888 $\pm$ 1.8e-3	0.4125 $\pm$ 1.1e-3
Our Method ($loss_{IPW}$)	0.6860 $\pm$ 1.6e-3	0.7079 $\pm$ 1.3e-3	0.7557 $\pm$ 1.2e-3	0.3784 $\pm$ 1.4e-3	0.3882 $\pm$ 1.4e-3	0.4119 $\pm$ 1.5e-3
DLA [2]	0.6892 $\pm$ 2.1e-3	0.7106 $\pm$ 2.2e-3	0.7578 $\pm$ 1.6e-3	0.3792 $\pm$ 2.7e-3	0.3870 $\pm$ 2.4e-3	0.4102 $\pm$ 2.9e-3
Regression EM [27]	0.6893 $\pm$ 2.1e-3	0.7108 $\pm$ 2.1e-3	0.7577 $\pm$ 1.6e-3	0.3707 $\pm$ 3.9e-3	0.3766 $\pm$ 3.7e-3	0.3977 $\pm$ 3.6e-3
Pairwise Debiasing [10]	0.6600 $\pm$ 3.5e-3	0.6847 $\pm$ 2.8e-3	0.7365 $\pm$ 2.0e-3	0.3481 $\pm$ 3.3e-3	0.3623 $\pm$ 2.8e-3	0.3912 $\pm$ 2.4e-3
Lower Bound (LambdaRank)	0.6541 $\pm$ 2.5e-3	0.6790 $\pm$ 2.2e-3	0.7320 $\pm$ 1.7e-3	0.3433 $\pm$ 2.2e-3	0.3579 $\pm$ 2.1e-3	0.3874 $\pm$ 1.7e-3

Table 3: Experiment Results on Categorical Labels

for binary classification ranker and pairwise ranker, and MSE loss for regression ranker. Besides, we tune the learning rate in [5e-3, 5e-2] carefully for each method. For a fair comparison, we train and evaluate each method 10 times and report the mean and standard deviation of evaluation metrics. To evaluate the unbiased LTR performance of our method, we adopt NDCG@k as the evaluation metric for experiments on categorical labels. As for continuous labels, we adopt the commonly used weighted summation function in typical recommendation applications to combine click and dwell-time, and define the evaluation metric $Reward@k$ as follows to simulate the combined live metric:

$$Reward@k = \sum_{i=1}^k click'_{u,x_i,i} + \frac{dt'_{u,x_i,i}}{e^\delta},$$

where $click'_{u,x_i,i}$ and $dt'_{u,x_i,i}$ are estimated click and dwell-time feedback at ranked position i . Unlike $click_{u,x_i,i}$, the position i here is determined by the final unbiased LTR ranker rather than the initial Ranking-SVM ranker. Specifically, $click'_{u,x_i,i}$ and $dt'_{u,x_i,i}$ are sampled based on Equation 21 and Equation 22 respectively.

5.3 Experiments on Continuous Labels (RQ1)

We first conduct experiments on the continuous synthesized label of Yahoo dataset and MovieLens to evaluate the performance of our method in real-world applications. For a comprehensive comparison, we train a pointwise DNN and LambdaRank model with the synthesized label without debiasing as the lower bound. Moreover, we remove the pairwise trust bias correction and loss refinement for direct metrics optimization from our method to explore their effectiveness. Note that the SOTA industrial recommender systems train multiple pointwise models to predict multiple objectives respectively and employ PAL [8] or sum-based shallow tower [31] for debiasing. We consider the following baseline methods:

- Sum-Synthesize: We train a ranker with the synthesized label and adopt the sum-based shallow tower for debiasing.
- PAL-Synthesize: We train a ranker with the synthesized label and adopt the PAL for debiasing.
- Sum-Click: We train the click and dwell-time task respectively and adopt sum-based shallow tower for debiasing on the click task.
- PAL-Click: We train the click and dwell-time task respectively and adopt PAL for debiasing on the click task.

- PAL-Click + PAL-DwellTime: We train the click and dwell-time task respectively and adopt PAL on both tasks.

For baseline models with multiple tasks, we combine the predicted click probability and dwell-time with the same weighted summation function as the synthesized label for ranking. As shown in Table 2, our method outperforms all baseline methods significantly in both datasets. One can see that LambdaRank performs worse than pointwise DNN with the biased synthesized label, which indicates that debiasing is more important for pairwise learning in recommendation scenarios and can explain why pointwise methods are the mainstream rankers for recommender systems before. Moreover, employing shallow tower on dwell-time harms the performance significantly, which shows that shallow tower cannot work well for continuous labels. In contrast, our probabilistic graphical based method has a clear separation of bias and relevance and achieves superior performance on debiasing and preference learning for different applications. Moreover, direct optimization for ranking metrics achieves significant improvement in both datasets. Besides, the pairwise trust bias correction exhibits better performance on Yahoo Dataset but similar performance on MovieLens. To explore why the pairwise trust bias correction performs differently on Yahoo Dataset and MovieLens, we analyze the statistics of m_{ij} defined in Equation 14 in these datasets and find that m_{ij} in different position pairs are roughly similar in MovieLens but differs a lot in Yahoo Dataset, which indicates that there is no improvement room for pairwise trust bias correction in MovieLens.

5.4 Experiments on Categorical Labels (RQ2)

We also perform experiments on the sampled binary click labels of Yahoo Dataset and Web30K to compare our method with the SOTA unbiased LTR methods including DLA [2], Regression EM [27] and Pairwise Debiasing [10] to answer RQ2. For a comprehensive comparison, we train LambdaRank models with the unbiased relevance labels and the biased click labels without debiasing as upper bound and lower bound respectively. It is shown in Table 3 that our method performs competitively with the SOTA unbiased LTR methods on categorical labels. Furthermore, the pairwise trust bias correction achieves consistent improvement on both Yahoo Dataset and Web30K. Besides, direct optimization for ranking metrics exhibits better performance on Web30K but similar performance on Yahoo Dataset as we can only compute ΔZ_{ij} based on the noisy

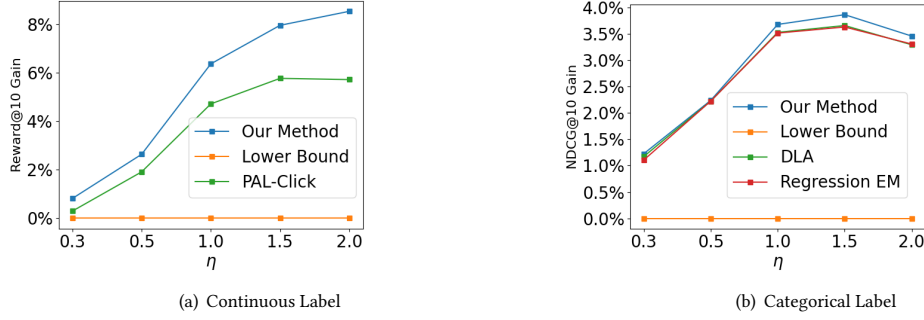


Figure 3: Performance Gain over Lower Bound with Different Severity of Position Bias

click signal. On the whole, our method achieves competitive performance on both continuous and categorical labels.

5.5 Robustness Analysis (RQ3)

To evaluate the robustness of the proposed method, we compare it with the best baseline methods, i.e. DLA, Regression EM and PAL-Click, on Yahoo Dataset under different severity of position bias and different weights of continuous labels in this section.

5.5.1 Severity of Position Bias. Figure 3 shows the performance gain of debiasing methods over the lower bound (LambdaRank) on continuous and categorical labels with different severity of position bias controlled by η . The performance gain is defined as the relative improvement on the evaluation metric over lower bound. One can see that our method is robust to the severity of position bias and outperforms other competitors consistently under different η .

5.5.2 Relative Weight of Continuous Label. According to Figure 4, our method performs best consistently under different weights of continuous labels controlled by δ . On the whole, our method is robust to different severity of position bias and different weights of continuous labels.

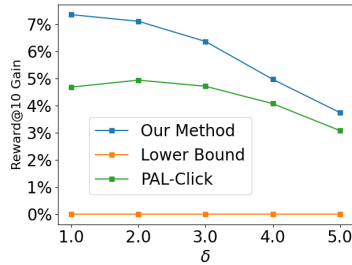


Figure 4: Performance Gain over Lower Bound with Different Relative Weights of Continuous Label

5.6 Online A/B Testing (RQ4)

To evaluate the performance of the proposed method in real-world systems, careful online A/B testing has been conducted in the ranking module of a large-scale video recommender system at Tencent News for 7 days. The live metric and synthesized label are defined as the combination of multiple user actions (e.g. click, dwell-time, liking, and sharing) in the system. As PAL-Click is the best performed baseline method on continuous labels, we train multiple user actions respectively and employ PAL on the click task for

debiasing as the baseline in online A/B testing. We implement all models in our deep learning framework, randomly distribute users into several buckets, and deploy each model to one of the buckets. Compared with PAL-Click, our method improves the live metric by 2.08%, which demonstrates the effectiveness of our method in debiasing and preference learning in real-world applications.

To further investigate the performance of our method in online A/B testing, we analyze the learned propensity per position denoted as θ_i of our method and compare it with the statistical probability of positive label in each position denoted as $P(c_i > 0)$ in the system. Figure 5 shows that θ_i is smaller for lower position, which is consistent with the decreasing trend of $P(c_i > 0)$. We also analyze the estimated probability of relevance at each position denoted as $\frac{P(c_i > 0)}{\theta_i}$ according to Equation 3. It is shown that $\frac{P(c_i > 0)}{\theta_i}$ also exhibits a stepwise decreasing trend especially in higher positions, which is also consistent with the recommendation mechanism that more relevant items is usually exposed in higher-ranked positions. In summary, our method can learn and correct position bias effectively in real-world recommendation applications.

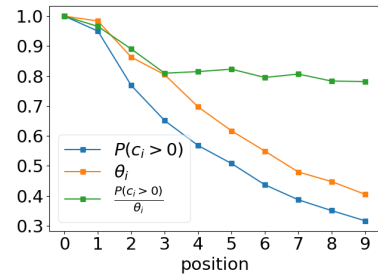


Figure 5: Statistical Probabilities and Learned Propensities

6 CONCLUSION

In this paper, we propose a novel unbiased pairwise LTR method to model position-based examination bias and trust bias in the pairwise fashion for better bias correction and preference learning, which is the first unbiased LTR method working for both categorical and continuous labels. Experiment results on public benchmark datasets and online A/B testing show that the proposed method achieves significant improvements over SOTA ranking models for continuous labels and competitive performance with SOTA unbiased LTR methods for categorical labels. Correcting for other types of bias besides position bias will be the focus of future work.

REFERENCES

- [1] Aman Agarwal, Xuanhui Wang, Cheng Li, Michael Bendersky, and Marc Najork. 2019. Addressing trust bias for unbiased learning-to-rank. In *The World Wide Web Conference*. 4–14.
- [2] Qingyao Ai, Keping Bi, Cheng Luo, Jiafeng Guo, and W Bruce Croft. 2018. Unbiased learning to rank with unbiased propensity estimation. In *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*. 385–394.
- [3] Christopher Burges, Robert Ragno, and Quoc Le. 2006. Learning to rank with nonsmooth cost functions. *Advances in neural information processing systems* 19 (2006), 193–200.
- [4] Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. 2005. Learning to rank using gradient descent. In *Proceedings of the 22nd international conference on Machine learning*. 89–96.
- [5] Christopher JC Burges. 2010. From ranknet to lambdarank to lambdamart: An overview. *Learning* 11, 23–581 (2010), 81.
- [6] Olivier Cappé and Eric Moulines. 2009. On-line expectation–maximization algorithm for latent data models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 71, 3 (2009), 593–613.
- [7] Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, and Xiangnan He. 2020. Bias and debias in recommender system: A survey and future directions. *arXiv preprint arXiv:2010.03240* (2020).
- [8] Huifeng Guo, Jinkai Yu, Qing Liu, Ruiming Tang, and Yuzhou Zhang. 2019. PAL: a position-bias aware learning framework for CTR prediction in live recommender systems. In *Proceedings of the 13th ACM Conference on Recommender Systems*. 452–456.
- [9] Yifan Hu, Yehuda Koren, and Chris Volinsky. 2008. Collaborative filtering for implicit feedback datasets. In *2008 Eighth IEEE International Conference on Data Mining*. Ieee, 263–272.
- [10] Ziniu Hu, Yang Wang, Qu Peng, and Hang Li. 2019. Unbiased LambdaMART: An unbiased pairwise learning-to-rank algorithm. In *The World Wide Web Conference*. 2830–2836.
- [11] Jui-Ting Huang, Ashish Sharma, Shuying Sun, Li Xia, David Zhang, Philip Pronin, Janani Padmanabhan, Giuseppe Ottaviano, and Linjun Yang. 2020. Embedding-based retrieval in facebook search. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2553–2561.
- [12] Kalervo Järvelin and Jaana Kekäläinen. 2002. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems (TOIS)* 20, 4 (2002), 422–446.
- [13] Kalervo Järvelin and Jaana Kekäläinen. 2017. IR evaluation methods for retrieving highly relevant documents. In *ACM SIGIR Forum*, Vol. 51. ACM New York, NY, USA, 243–250.
- [14] Thorsten Joachims, Laura Granka, Bing Pan, Helene Hembrooke, and Geri Gay. 2017. Accurately interpreting clickthrough data as implicit feedback. In *ACM SIGIR Forum*, Vol. 51. Acm New York, NY, USA, 4–11.
- [15] Thorsten Joachims, Adith Swaminathan, and Tobias Schnabel. 2017. Unbiased learning-to-rank with biased feedback. In *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining*. 781–789.
- [16] Christopher C Johnson. 2014. Logistic matrix factorization for implicit feedback data. *Advances in Neural Information Processing Systems* 27, 78 (2014), 1–9.
- [17] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix factorization techniques for recommender systems. *Computer* 42, 8 (2009), 30–37.
- [18] Jae-won Lee, Seongmin Park, and Jongwuk Lee. 2021. Dual Unbiased Recommender Learning for Implicit Feedback. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1647–1651.
- [19] Dawen Liang, Laurent Charlin, James McInerney, and David M Blei. 2016. Modeling user exposure in recommendation. In *Proceedings of the 25th international conference on World Wide Web*. 951–961.
- [20] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2012. BPR: Bayesian personalized ranking from implicit feedback. *arXiv preprint arXiv:1205.2618* (2012).
- [21] Douglas A Reynolds. 2009. Gaussian mixture models. *Encyclopedia of biometrics* 741, 659–663 (2009).
- [22] Matthew Richardson, Ewa Dominowska, and Robert Ragno. 2007. Predicting clicks: estimating the click-through rate for new ads. In *Proceedings of the 16th international conference on World Wide Web*. 521–530.
- [23] Yuta Saito. 2020. Unbiased Pairwise Learning from Biased Implicit Feedback. In *Proceedings of the 2020 ACM SIGIR on International Conference on Theory of Information Retrieval*. 5–12.
- [24] Yuta Saito, Suguru Yaginuma, Yuta Nishino, Hayato Sakata, and Kazuhide Nakata. 2020. Unbiased recommender learning from missing-not-at-random implicit feedback. In *Proceedings of the 13th International Conference on Web Search and Data Mining*. 501–509.
- [25] Hongyan Tang, Junling Liu, Ming Zhao, and Xudong Gong. 2020. Progressive layered extraction (ple): A novel multi-task learning (mtl) model for personalized recommendations. In *Fourteenth ACM Conference on Recommender Systems*. 269–278.
- [26] Xuanhui Wang, Michael Bendersky, Donald Metzler, and Marc Najork. 2016. Learning to rank with selection bias in personal search. In *Proceedings of the 39th International ACM SIGIR conference on Research and Development in Information Retrieval*. 115–124.
- [27] Xuanhui Wang, Nadav Golbandi, Michael Bendersky, Donald Metzler, and Marc Najork. 2018. Position bias estimation for unbiased learning to rank in personal search. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*. 610–618.
- [28] Xuanhui Wang, Cheng Li, Nadav Golbandi, Michael Bendersky, and Marc Najork. 2018. The lambdaloss framework for ranking metric optimization. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management*. 1313–1322.
- [29] Xinwei Wu, Hechang Chen, Jiashu Zhao, Li He, Dawei Yin, and Yi Chang. 2021. Unbiased learning to rank in feeds recommendation. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*. 490–498.
- [30] Peifeng Yin, Ping Luo, Wang-Chien Lee, and Min Wang. 2013. Silence is also evidence: interpreting dwell time for recommendation from psychological perspective. In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*. 989–997.
- [31] Zhe Zhao, Lichan Hong, Li Wei, Jilin Chen, Aniruddh Nath, Shawn Andrews, Aditee Kumbhakar, Maheswaran Sathiamoorthy, Xinyang Yi, and Ed Chi. 2019. Recommending what video to watch next: a multitask ranking system. In *Proceedings of the 13th ACM Conference on Recommender Systems*. 43–51.